

Cyber Security Internship Report

Student Name: RUDRESHA RK

University: Presidency University Bangalore Karnataka

Company Name: Anonymous India Cyber Defense Academy

Major: Computer Science Engineering (Cybersecurity)

Duration: April 2, 2026 to May 2, 2026

Domain: Ethical Hacking

Mentor: Akash Das

Coordinator: Akash Das

Projects and Tasks (Level-Beginner)

Assessment 2: Secure Lab Configuration for Penetration Testing

1. Introduction

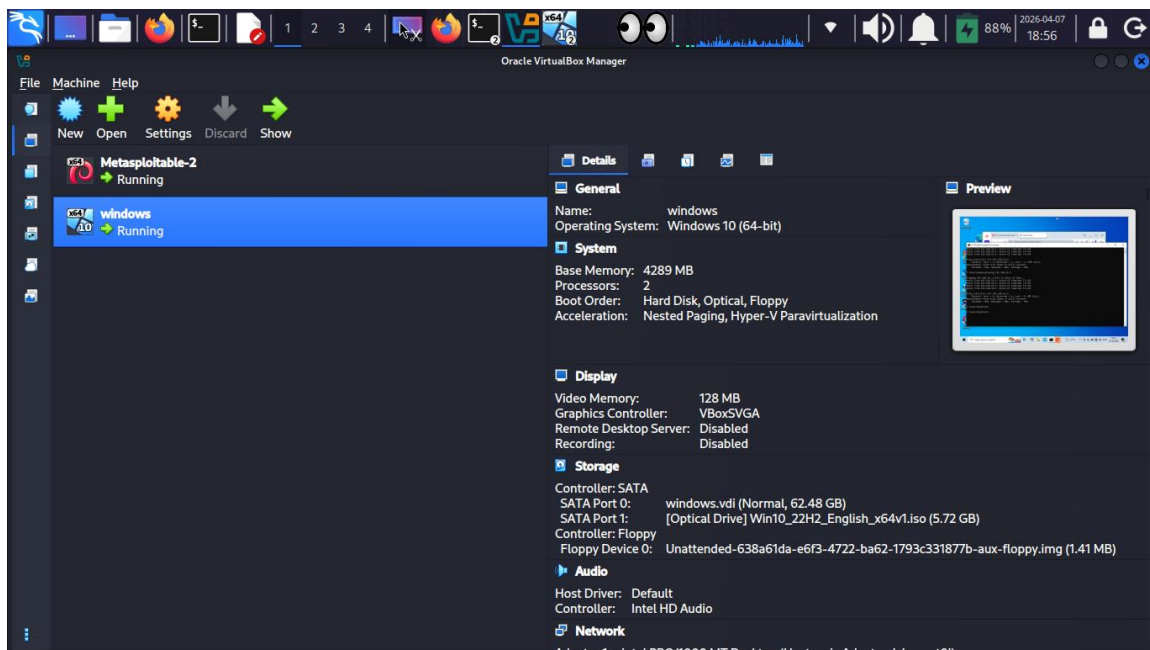
The objective of this task was to build a secure, isolated "sandbox" environment to practice penetration testing techniques without risking the host system or external networks. By using virtualization, I can simulate real-world attacks in a controlled space where all traffic is contained.

2. Lab Infrastructure & Specifications

I chose Oracle VM VirtualBox as the primary hypervisor for this setup due to its robust snapshot management and networking features. The lab consists of three distinct virtual machines (VMs):

- **Attacker Machine (Kali Linux):** The primary OS for security auditing, configured with 2GB RAM and 40GB storage.
- **Vulnerable Target (Metasploitable 2):** A Linux-based VM intentionally configured with multiple security holes to practice exploitation.
- **Windows Vulnerable Target (Windows 10):** A common client OS used to test how standard security features (like Windows Defender) respond to unauthorized access and to practice Windows-specific exploitation.

Output:



3. Step-by-Step Implementation

3.1 Installation and Configuration:

For this setup, I am utilizing **Kali Linux as my primary (Host) Operating System**, providing a native environment with full access to hardware resources for security auditing. To create a safe testing ground, I installed **Oracle VM VirtualBox** directly on my Kali host to manage the target machines.

3.2 Network Isolation (Host-Only Adapter)

This was the most critical step to ensure safety. I configured all three VMs to use a **Host-only Network** adapter.

- **Why:** This creates a private internal network where the VMs can communicate with each other but have no path to the physical internet or my home Wi-Fi.

IP Addressing: Each machine was assigned a static IP within the same subnet (e.g., 192.168.56.0/24) to ensure stable communication during tests.

3.3 Target IP Table:

Machine	OS	IP Address	Role
HOST	KALI LINUX	192.168.56.1	Attacker
VM1	Metasploitable-2	192.168.56.101	Linux Target
VM2	Windows-10	192.168.56.102	Windows Target

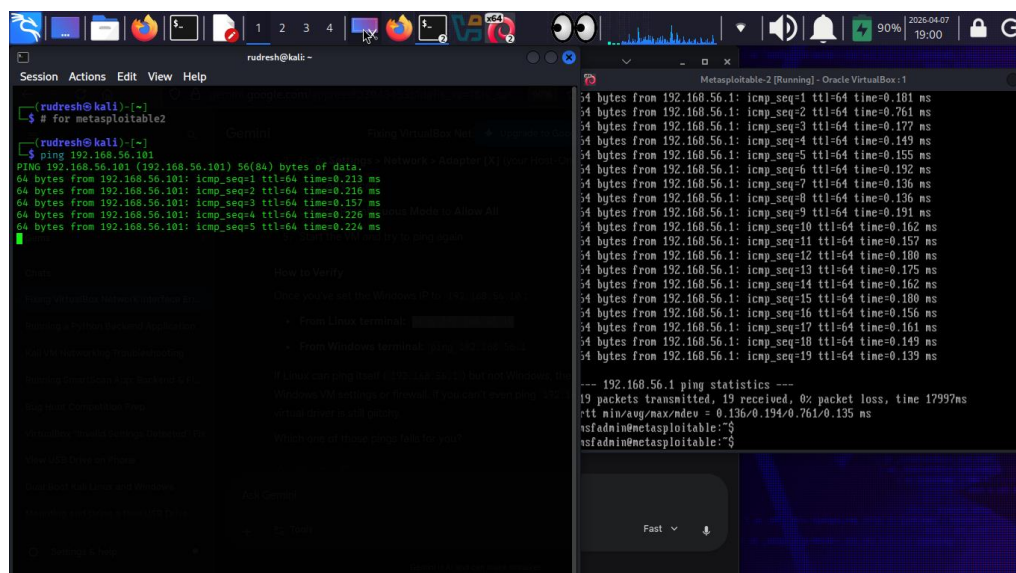
Personal Lab Note: "The most challenging part was ensuring the Windows 10 firewall didn't block my pings. I learned that in a real-world assessment, a 'no ping' response doesn't always mean the host is down—it often just means the ICMP traffic is being filtered."

3.4 Connectivity Testing

To verify the setup, I performed a connectivity test from the Kali Linux terminal:

- **Command:** ping -c 4 192.168.56.101
- Testing metasploitable2 machine connectivity

Output:



```
(rudresh@kali) ~  
$ # for metasploitable2  
(rudresh@kali) ~  
$ ping 192.168.56.101  
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:  
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=0.213 ms  
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=0.216 ms  
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=0.157 ms  
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=0.226 ms  
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=0.224 ms  
^C  
--- 192.168.56.101 ping statistics ---  
5 packets transmitted, 5 received, 0% packet loss, time 1799ms  
rtt min/avg/max/mdev = 0.136/0.194/0.761/0.135 ms  
nsfadmin@metasploitable:~$  
nsfadmin@metasploitable:~$
```

-
- **Result:** The test showed 0% packet loss, confirming that the "Attacker" machine can successfully see and interact with the "Target" machines within the isolated environment.
- **Command:** ping -c 4 192.168.56.102
- Testing Windows-10 machine connectivity
- Output:

The screenshot shows a Kali Linux terminal window on the left and a Windows 10 virtual machine window on the right. The terminal window displays the output of a ping command from a Kali machine to a Windows machine at IP 192.168.56.102. The output shows 16 successful ping requests with 0% packet loss. The Windows 10 window shows the desktop with icons for Recycle Bin, Profile 2 - Edge, Google Chrome, Microsoft Edge, NetScanTools Basic Edition, and NeoTracePro. The taskbar at the bottom of the Windows window shows the Start button and several pinned applications.

```

(rudresh@kali)-[~]
$ #for windows ip
(rudresh@kali)-[~]
$ ping 192.168.56.102
PING 192.168.56.102 (192.168.56.102) 56(84) bytes of data:
64 bytes from 192.168.56.102: icmp_seq=1 ttl=128 time=0.403 ms
64 bytes from 192.168.56.102: icmp_seq=2 ttl=128 time=0.361 ms
64 bytes from 192.168.56.102: icmp_seq=3 ttl=128 time=0.327 ms
64 bytes from 192.168.56.102: icmp_seq=4 ttl=128 time=0.335 ms
64 bytes from 192.168.56.102: icmp_seq=5 ttl=128 time=0.402 ms
64 bytes from 192.168.56.102: icmp_seq=6 ttl=128 time=0.837 ms
64 bytes from 192.168.56.102: icmp_seq=7 ttl=128 time=0.256 ms
64 bytes from 192.168.56.102: icmp_seq=8 ttl=128 time=0.436 ms
64 bytes from 192.168.56.102: icmp_seq=9 ttl=128 time=0.276 ms
64 bytes from 192.168.56.102: icmp_seq=10 ttl=128 time=0.293 ms
64 bytes from 192.168.56.102: icmp_seq=11 ttl=128 time=0.305 ms
64 bytes from 192.168.56.102: icmp_seq=12 ttl=128 time=0.273 ms
64 bytes from 192.168.56.102: icmp_seq=13 ttl=128 time=0.246 ms
64 bytes from 192.168.56.102: icmp_seq=14 ttl=128 time=0.301 ms
64 bytes from 192.168.56.102: icmp_seq=15 ttl=128 time=0.304 ms
64 bytes from 192.168.56.102: icmp_seq=16 ttl=128 time=0.307 ms
^C

```

- **Result:** The test showed 0% packet loss, confirming that the "Attacker" machine can successfully see and interact with the "Target" machines within the isolated environment.

3.5 System Recovery (Snapshots)

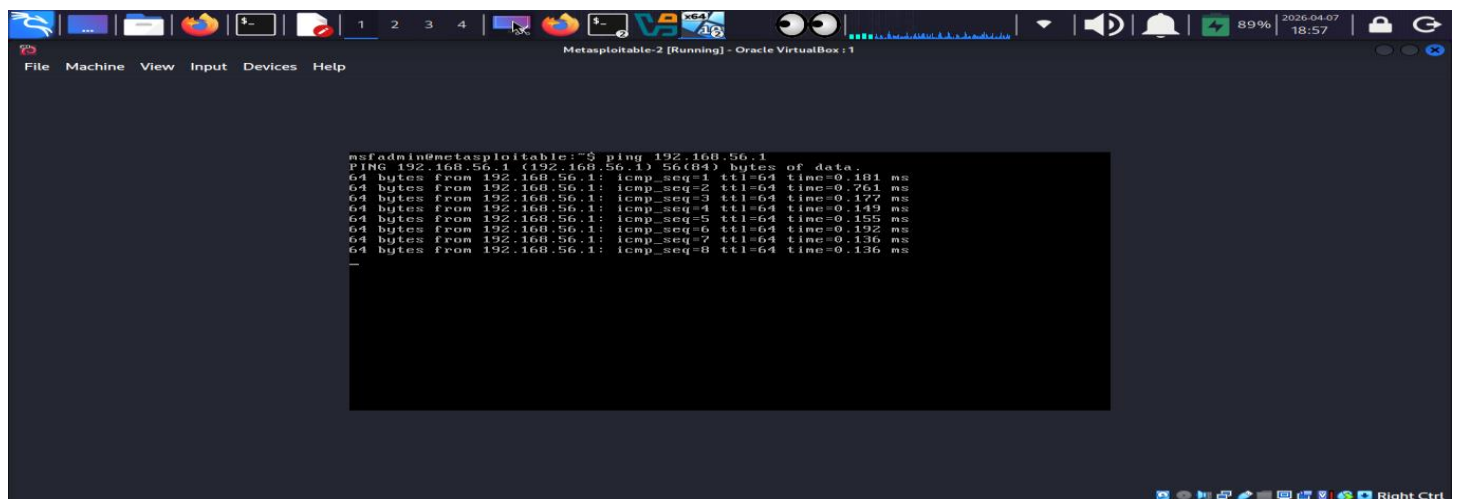
After confirming connectivity, I took a **Snapshot** of each VM in its "Clean State."

- **Outcome:** If a future exploit causes a system crash or deletes critical files on the target, I can restore the machine to this exact point in seconds, preventing the need for a full re-installation.

Snapshots:

Kali linux ip: 192.168.56.1

IN Metasploitable-2 Use The Command: ping 192.168.56.1

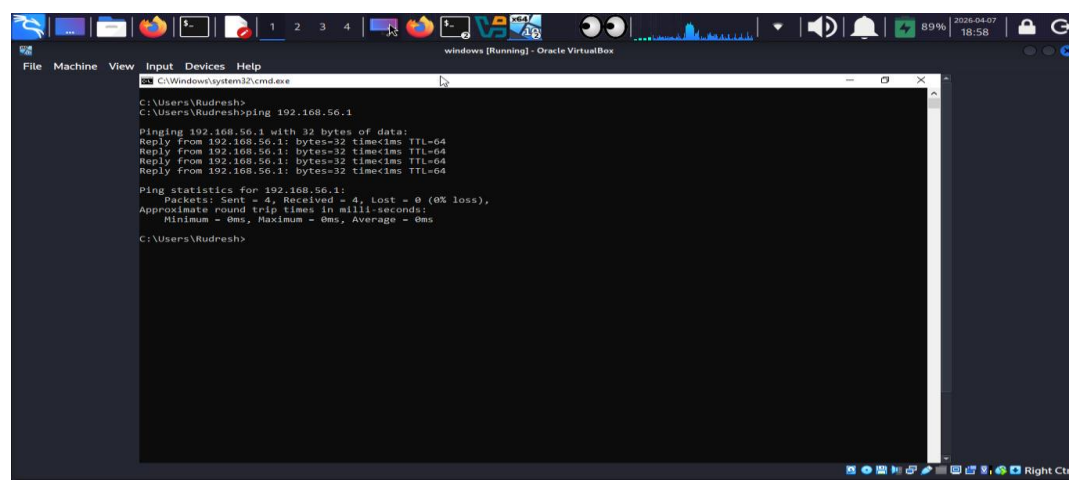


```
msfadmin@metasploitable:~$ ping 192.168.56.1
PING 192.168.56.1 (192.168.56.1) 56(84) bytes of data:
64 bytes from 192.168.56.1: icmp_seq=1 ttl=64 time=0.101 ms
64 bytes from 192.168.56.1: icmp_seq=2 ttl=64 time=0.761 ms
64 bytes from 192.168.56.1: icmp_seq=3 ttl=64 time=0.177 ms
64 bytes from 192.168.56.1: icmp_seq=4 ttl=64 time=0.149 ms
64 bytes from 192.168.56.1: icmp_seq=5 ttl=64 time=0.155 ms
64 bytes from 192.168.56.1: icmp_seq=6 ttl=64 time=0.192 ms
64 bytes from 192.168.56.1: icmp_seq=7 ttl=64 time=0.136 ms
64 bytes from 192.168.56.1: icmp_seq=8 ttl=64 time=0.136 ms
^C
```

//Metasploitable-2 successfully pinging to kali linux ip address

//Windows-10 successfully pinging to kali linux ip address

IN Windows Terminal use the command :ping 192.168.56.1



```
C:\Windows\system32\cmd.exe
C:\Users\Rudresh>
C:\Users\Rudresh>ping 192.168.56.1

Pinging 192.168.56.1 with 32 bytes of data:
Reply from 192.168.56.1: bytes=32 time=1ms TTL=64
Reply from 192.168.56.1: bytes=32 time=1ms TTL=64
Reply from 192.168.56.1: bytes=32 time=1ms TTL=64
Reply from 192.168.56.1: bytes=32 time=1ms TTL=64

Ping statistics for 192.168.56.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Rudresh>
```

Outcome: Successfully configured All the machines. All are in same network and successfully communicating each other and ready for Attacking Simulation

4. Skills Acquired

- **Virtualization Management:** Proficiency in configuring hardware resources (RAM/CPU) for multiple guest operating systems.
- **Network Engineering:** Understanding of internal vs. external networking and the importance of isolation in security testing.
- **System Hardening:** Learning the value of snapshots for disaster recovery and persistent testing environments.

5. Conclusion

Setting up this lab has provided me with a safe platform to apply the theoretical knowledge gained in Assessment 1. I now have the infrastructure ready to perform port scanning, directory brute-forcing, and traffic analysis tasks as part of my practical training.

Outcome:

By configuring Kali Linux as my primary OS, I built a high-performance attacking platform with direct hardware access to interact with isolated virtual targets. This setup taught me to manage host-to-guest communication through "Host-only" networking, ensuring a safe environment for exploitation practice. I now have a

functional, snapshot-protected laboratory that allows for repeatable testing against both Linux and Windows environments.